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RESEARCH

Investigating nozzle-based additive manufacturing dynamics with high-speed imaging and machine learning

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Abstract
Additive manufacturing (AM) offers great design flexibility but remains constrained by the complexity of tuning processing parameters from a vast parameter search space either by traditional trial-and-error experimental approach or expensive computational fluid dynamics simulations. Further, AM processes involve critical transient phenomena at very small timescales ranging from microsecond to millisecond, which are difficult to observe and characterize experimentally. This work presents a novel approach that combines high-speed photography with machine learning (ML) to model and predict dynamic printing behavior. We used jet printing data obtained from Electrohydrodynamic (EHD) jet printing as a representative AM process. A Gaussian Process (GP) classifier is trained on 123 high-speed video sequences captured at 15,000 frames per second accurately predicts five printing regimes, including cone-jetting, micro-dripping, dripping, unstable, and no-print, across a wide range of parameters. Further, a deep learning model combining a 3D convolutional encoder and transformer-based temporal predictor provided highly accurate quantitative estimation of frame-by-frame droplet dynamics at a computational cost orders of magnitude less than equivalent computational fluid dynamics simulations. These results establish a data-driven framework for the simulation, predictive control, and rapid optimization of EHD printing dynamics, demonstrating the potential of high-speed imaging-informed machine learning as a computationally efficient surrogate for complex AM processes.

Keywords Machine learning (ML) · Additive manufacturing (AM) · Artificial intelligence (AI) · High-speed imaging · Electrohydrodynamic (EHD) jet printing · Transformer neural networks

1 Introduction

Additive manufacturing (AM) has emerged as a time-efficient method for producing cost-effective, arbitrarily complex 3D geometries [1]. Unlike traditional subtractive or formative manufacturing processes, AM achieves high geometric complexity without substantial increases in manufacturing costs or production timelines, making it especially valuable in fields demanding customized components. Additive manufacturing techniques have been developed to create rapid prototypes and production-scale parts across a wide range of applications and are now increasingly utilized in full-scale manufacturing across several domains, including aerospace components, biomedical implants, and integrated circuits [2–4]. This versatility is further supported by advancements that enable the processing of numerous materials, ranging from conventional polymers and metallic alloys to emerging classes such as composites and nanomaterials [5].

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Despite the benefits offered by AM technologies, their implementation and optimization are substantially hindered by the complexity of parameter tuning. Most of the AM processes are governed by a large number (>15) of processing parameters including printer settings, material properties, and environmental conditions. Even under conservative assumptions, such as evaluating only a few discrete values per parameter, the number of possible parameter combinations grows exponentially. The size of the parameter space renders exhaustive experimentation expensive and impractical and presents a major barrier to systematically achieving optimal AM performance.

This optimization challenge is further compounded by the fact that most AM processes involve rapid, transient phenomena that occur on microsecond to millisecond timescales. These events are critical to process outcomes, but they are difficult to observe and characterize experimentally. Every AM process exhibits unique multi-physics behaviors. For example, laser powder bed fusion involves melt pool dynamics and spatter formation within hundreds of microseconds [6]. Fused deposition modeling exhibits polymer flow and resolidification during millisecond-scale extrudate deposition [7]. The fundamental mechanisms governing print quality operate at temporal scales exceeding conventional monitoring capabilities. Traditional process development approaches rely primarily on post-processing analysis of printed components [8]. This methodology provides limited insight into the in-situ dynamic behaviors determining these outcomes.

Given these observational limitations and the complexity of rapid multi-physics phenomena, understanding the interactions between all parameters and their effects on the printing process is vital for achieving a stable process. Finite element analysis (FEA) and computational fluid dynamics (CFD) modeling based methods have been developed to simulate various AM techniques [9–12]. However, due to the complex multi-physics behaviors inherent in most AM technologies, most of the models are limited to a few processing parameters, which makes them difficult to apply to generic printing cases [13]. Furthermore, computational simulation approaches typically rely on solving sets of partial differential equations (PDEs) such as the Navier–Stokes equations describing fluid flow [14]. Solving these PDEs with sufficient fidelity and resolution is computationally intensive, limiting scalability, speed, and rendering online process control infeasible when the parameter space is large [15].

As such, there exists a demonstrable need for approaches capable of efficiently exploring the vast parameter space, accurately and quickly simulating dynamic process behaviors, and establishing a foundation for process control in multi-physics systems. Such approaches should ideally combine experimental insight with computational efficiency, providing researchers with valuable insight into the underlying transient physical phenomena at short time scales while enabling rapid optimization and simulation of AM processes.

In this work, we navigate a representative multi-physics AM process by investigating electrohydrodynamic (EHD) jet printing, which is governed by complex multi-physics processes that are difficult to explore experimentally. EHD printing is a high-precision nozzle-based AM technique that relies on electric fields to drive the deposition of fluid materials at micro-/nano-scale resolutions [16]. In EHD printing, a conductive nozzle connected to a fluid reservoir is positioned above a grounded substrate. By applying a sufficiently high electric potential difference between the nozzle and the substrate, a strong electric field is generated. This electric field exerts electrostatic forces that deform the fluid meniscus into a conical shape known as the Taylor cone [17]. When the electrostatic stress overcomes the combination of viscoelastic and surface tension stresses, a drop or jet is formed. EHD printing is governed by several factors, such as system parameters, process parameters, environmental conditions, and material properties [18–20]. Depending on the specific set of those parameters, distinct modes such as micro-dripping, dripping, or cone-jetting can emerge [21–23]. The physics governing these modes and their transitions have been characterized in foundational electrospray and EHD literature [17, 21–23]. The present work builds upon this established knowledge by applying high-speed imaging and machine learning to systematically classify, map, and simulate these behaviors across the process parameter space. These modes are characterized by different formation mechanisms, and each yields different levels of stability, jet breakup, and droplet detachment characteristics. Each of these dynamic behaviors has substantial implications for print quality, resolution, and repeatability. The five modes, illustrated in Sect. 3.1, are as follows [20]:

1. *Cone-jetting:* A steady-state regime in which a stable Taylor cone forms and emits a continuous jet of fluid (filament), yielding high-resolution, precision-controlled deposition
2. *Micro-dripping:* A periodic mode characterized by the formation of small, regularly spaced droplets emitted at moderate frequencies, typically under moderate field strength
3. *Dripping:* A higher-energy mode in which larger droplets are emitted, often chaotically, and typically driven by excessive voltage or high pressure
4. *Unstable:* A regime exhibiting erratic or multi-modal behavior, including non-axial or multi-jetting, generally

caused by a poor balance between pressure, electric field, and shear forces

5. *No-print:* A regime in which no visible jet or droplet emission occurs, corresponding to insufficient electrostatic force or pressure to drive fluid flow

In addition, the stability, size, and shape of this cone directly influence printing characteristics, including droplet formation, filament diameter, and the overall quality and precision of the printed structures [24]. Various computational approaches have been developed to simulate EHD printing phenomena [25–29], but these models are typically limited to specific materials or printing modes due to the complex multi-physics nature of the process.

A critical step toward addressing the challenges of current modeling and simulation approaches involves obtaining high-fidelity, temporally resolved experimental data characterizing the dynamic behaviors of printing processes. Fundamental phenomena, including fluid-ejection, droplet or filament formation and separation, jet breakup, and mode transitions, frequently occur on sub-millisecond timescales [30–33]. These transient events significantly impact the quality, precision, and repeatability of printed structures but remain virtually impossible to observe when utilizing conventional in-situ imaging methods, which typically offer insufficient temporal resolution [34, 35].

High-speed photography has emerged as an essential tool capable of overcoming these limitations by offering both the frame rate and spatial resolution required to capture rapidly changing printing dynamics. It offers the temporal and spatial resolution necessary to capture rapid flow phenomena that influence material distribution and process stability and repeatability. It has been used as an essential analysis tool for Laser Powder Bed Fusion (L-PBF) [36–38] and as a qualitative visualization tool for nozzle-based AM techniques [16]. However, its application in understanding fluid dynamics at the nozzle, particularly in terms of droplet formation, jet breakup, and flow instabilities during material extrusion and deposition remains comparatively limited.

On the other hand, machine learning (ML) methods have become increasingly valuable for addressing practical challenges encountered in AM, particularly regarding the optimization, simulation, and predictive control of complex, high-dimensional parameter spaces [39, 40]. ML encompasses computational techniques capable of learning relationships directly from data, enabling rapid predictions of process outcomes without the necessity of analytical or numerical methods. Unlike traditional physics-based simulation methods, which are computationally expensive and often limited in scalability and flexibility, ML algorithms leverage statistical inference and pattern recognition techniques to efficiently generalize from observed data to unexplored operational regimes. In material-jetting processes, ML has been applied to inkjet printing for in-situ droplet monitoring and process classification using image analysis [35], as well as to study jetting behavior in drop-on-demand systems [30]. Due to the complex, multi-physics nature of EHD printing, data-driven ML modeling techniques have also been pursued, including soft-computing approaches for performance prediction [41], physics-informed Bayesian learning of jetting dynamics [42], and broader machine learning frameworks for parameter analysis and feature characterization [43–45]. However, these prior ML studies for EHD printing were primarily developed using large numerical simulation datasets or post-process feature measurements, and do not leverage temporally resolved high-speed experimental observations. To the best of our knowledge, applying ML to investigate these rapid EHD phenomena using temporally resolved high-speed experimental data has not been explored. Additionally, data-driven approaches have not been utilized to develop ML-based high-fidelity dynamic simulation surrogates grounded in high-speed imaging for EHD printing or AM processes more broadly. The present work directly addresses this gap by integrating high-speed photography with ML to enable both regime classification and dynamic simulation of EHD printing, explicitly distinguishing it from and advancing beyond prior data-driven efforts in this domain.

The unified research question is: Can experimental high-speed imaging data, without reliance on physics-based simulation, serve as the foundation for understanding the process-structure relationship in additive manufacturing by informing surrogate models for both regime-level classification and dynamic simulation of EHD printing? To answer this question, two complementary ML approaches are employed that together form a comprehensive data-driven framework. First, a Gaussian Process (GP) classifier maps the voltage-pressure process parameter space to discrete printing regimes, providing probabilistic process characterization that guides parameter selection. Second, a deep learning surrogate model predicts the temporal evolution of droplet dynamics within each regime, replacing computationally intensive CFD simulations with fast, data-driven inference. Together, regime identification from the GP classifier informs and contextualizes the dynamic predictions of the deep learning model, and both are grounded in the same high-speed experimental dataset, forming a coherent, integrated framework that informs researchers of the process-structure relationship. High-speed photography is performed to visualize the fluid-flow phenomena occurring during EHD printing. Crucial physical phenomena, including the formation and stability of electrohydrodynamic Taylor cones, transitions between dripping and jetting modes, and instability phenomena, are observed and systematically

quantified through high-speed imaging. By capturing EHD printing dynamics at microsecond resolution, the research enables rapid exploration of complex parameter spaces that would be prohibitively expensive using traditional CFD/FEA methods. The resulting ML surrogates provide orders of magnitude faster predictions than physics-based approaches, establishing a practical foundation for predictive EHD printing process optimization and control across multiple printing regimes and dynamic behaviors.

2 Methodology

2.1 Experimental setup

The schematic of the EHD printing system is depicted in Fig. 1. The primary components of the system include a conductive nozzle, a grounded ring electrode, a high-voltage power supply, a printing substrate, and a translating stage, as well as the high-speed camera apparatus, consisting of the camera, lens, and associated mounting and lighting fixtures.

The printing fluid is supplied to the conductive nozzle via a pump system (labeled as P in Fig. 1), which controls the flow rate and delivery of fluid material to the nozzle tip. The fluid material utilized in this experimentation was glycerin dyed to provide sufficient contrast for high-speed imaging data. A high-voltage power supply is employed to apply an electrical potential difference ΔV between the conductive nozzle and the grounded ring electrode. The substrate, positioned beneath the nozzle, is mounted on a translating stage capable of 3-axis motion, allowing for accurate deposition and patterning of fluid materials according to the desired geometry.

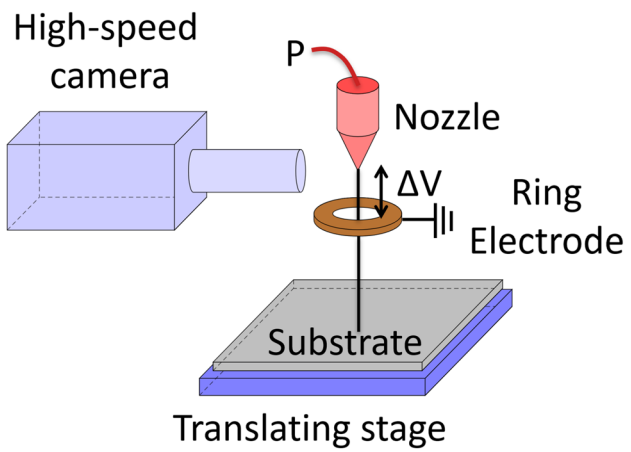


Fig. 1 The experimental setup utilized in this study to capture images of the EHD jetting process at the fluid meniscus. A potential applied across a ring electrode and the nozzle tip drives fluid deposition on the substrate

To effectively capture the dynamic phenomena occurring during the printing process, a high-speed camera system is integrated into the experimental setup. The camera system is oriented perpendicular to the nozzle. The camera used was the IDT MotionPro Y4-S2 high-speed camera, mounted with a microscope reflex lens, which achieved a resolution of 4.43 μm/pixel. High-speed images were recorded at up to 15,000 frames per second, providing temporal resolution sufficient to adequately analyze sub-millisecond events without considerable motion blur.

Video sequences were captured at various voltage and pressure settings with a fixed standoff distance (ring electrode to nozzle) of 4 mm. The range of voltages tested at was 2800–6000 V, which contains all of the typically observed printing modes for glycerin in EHD [20]. The range of pressures tested at was 0 psi to 5 psi. The (V, P) combinations were randomly selected and distributed relatively uniformly within the sample space. The voltage and pressure were initialized and the printing behavior was allowed to reach steady state by a 10-second delay. The lights were then activated and the camera set for 200 frames to record printing phenomena (e.g., cone formation). Overall, 123 videos at various pressure and voltage settings were added to the machine learning dataset, and 219 additional videos were captured of substrate-nozzle interactions.

2.2 Video sequence analysis

The video sequences were analyzed using Xcitex ProAnalyst motion analysis software, which provides frame-by-frame tracking of user-defined image features to extract quantitative, time-resolved measurements from high-speed photography. This image-based tracking approach is well suited for liquid-phase imaging [46].

The analysis was applied exclusively to the dripping and micro-dripping printing regimes, where the droplet and meniscus boundaries are clearly distinguishable from the background in the grayscale images. In these modes, the trackable feature was the leading edge (frontier) of the forming droplet, which provides a well-defined, high-contrast boundary that ProAnalyst reliably identifies across successive frames; contrast was further enhanced by the use of dyed glycerin as the printing fluid (see Sect. 2.1). The region of interest (ROI) was defined to encompass the nozzle tip and the droplet formation zone directly below it, capturing the full extent of meniscus elongation, necking, and pinch-off during each ejection cycle. The nozzle outer diameter (208 μm), which is clearly visible in each frame, served as the calibration length to convert pixel coordinates to physical dimensions. Droplet frontier position was extracted at each frame, and velocity was computed via finite differences over consecutive frames, yielding the velocity at

the moment of jet breakage. Measurements were repeated across multiple ejection cycles within each video sequence to assess cycle-to-cycle consistency.

The cone-jetting regime was excluded from this tracking analysis because, in steady-state cone jetting, there is no discrete droplet frontier to track: the process produces a continuous, opaque filament rather than individual forming droplets, and the frontier-based tracking method is therefore inapplicable. The unstable regime was similarly excluded due to its erratic, multi-modal jetting behavior.

By pairing high-speed photography with this tracking approach, this study enables precise spatiotemporal quantification of fluid behaviors that are otherwise invisible in real-time observation. The DIC-derived measurements inform the machine learning models discussed in Sect. 2.3: tracked droplet contours serve as ground truth target sequences for the deep learning prediction model, and the jet breakage velocity measurements provide training data for the Gaussian Process regression model used to generate the velocity process map in Sect. 3.4.

2.3 Machine learning models

2.3.1 Gaussian processes for process mapping

High-speed video data acquisition allowed for a visual inspection and frame-by-frame analysis of EHD printing phenomena, enabling identification and manual labeling of five distinct printing modes, which were subsequently used as categorical labels for supervised learning.

To achieve accurate classification of the five printing regimes and generate a processing map, a Gaussian Process (GP) classifier was employed due to its ability to model complex, non-linear decision boundaries while also providing inherent uncertainty quantification.

Gaussian Processes are class of non-parametric, Bayesian machine learning models that define a prior over functions and perform inference by conditioning this prior on observed data. In the context of this study, each input to the GP classifier is a two-dimensional vector $\mathrm{x} = [V, P]^T$ consisting of the applied voltage V and pressure P. Each data point is associated with a discrete class label $y \in \{1, 2, 3, 4, 5\}$ corresponding to one of the five printing modes: cone-jetting, micro-dripping, dripping, unstable, or no-print.

The Gaussian Process assumes that the latent function values $f(\mathrm{x})$ at any finite collection of inputs $\mathrm{x}_1, \ldots, \mathrm{x}_n$ follow a joint multivariate Gaussian distribution:

$$f(\mathrm{x}) \sim \mathcal{GP}(m(\mathrm{x}), k(\mathrm{x}, \mathrm{x}')) \tag{1}$$

Here, $m(\mathrm{x})$ is the mean function, typically set to zero, and $k(\mathrm{x}, \mathrm{x}')$ is the covariance function (or kernel) that encodes assumptions about the smoothness of functions and their correlation across the input space.

In this study, the radial basis function (RBF) was used:

$$k(\mathrm{x}, \mathrm{x}') = \sigma_f^2 \exp\left(-\frac{\|\mathrm{x} - \mathrm{x}'\|^2}{2\ell^2}\right) \tag{2}$$

where σ_f^2 is the signal variance and ℓ is the length scale hyperparameter controlling the sensitivity to changes in the input space. This kernel is commonly used and was selected for its flexibility in modeling nonlinear class boundaries.

For classification tasks, the latent GP output is passed through a softmax function to yield class probabilities. That is, for a C-class problem (here, $C = 5$), the predictive probability distribution for an input x_* is given by:

$$p(y_* = c|\mathrm{x}_*) = \frac{exp(f_c(\mathrm{x}_*))}{\sum_{c'=1}^{C} exp(f_{c'}(\mathrm{x}_*))} \tag{3}$$

where $f_c(\mathrm{x}_*)$ is the latent GP function corresponding to class c evaluated at the test point x_*. In practice, approximate inference methods are used to estimate posterior class probabilities, as exact inference is generally intractable in GP classification, especially for larger datasets. A more complete description of the implementation of Gaussian Processes for machine learning applications can be found in the literature [47, 48].

A critical strength of the GP classifier is its probabilistic nature, as it achieves state-of-the-art estimation of epistemic uncertainty. Unlike deterministic classifiers that output a single hard label, GP classification outputs a distribution over class labels for each input point, reflecting both the prediction and its associated confidence. This enables the construction of not just a process map indicating the most likely printing mode across the voltage-pressure domain, but also a corresponding uncertainty map indicating where predictions are more or less certain. This uncertainty is particularly valuable in experimental systems where data scarcity or high stochasticity can make deterministic models brittle or overconfident. Critically, the predicted class boundaries correspond directly to regions of highest predictive uncertainty: at a boundary, the posterior probabilities of adjacent classes approach equality, producing maximum entropy in the predictive distribution. Physically ambiguous behavior is expected and observed at these boundaries. They represent transitional parameter conditions where the printing system does not exhibit a single stable mode, validating the alignment between the model's uncertainty and the underlying physical reality.

The 123-video dataset is distributed across the five printing modes as follows: no-print ($n = 4$), unstable ($n = 9$), micro-dripping ($n = 46$), cone-jetting ($n = 17$), and dripping

($n = 50$). This class imbalance reflects the physical reality of the parameter space: dripping and micro-dripping dominate larger regions of the voltage-pressure domain, while cone-jetting, unstable, and no-print modes occupy narrower operating windows. Experimental conditions were sampled randomly and distributed approximately uniformly across the voltage-pressure parameter space to ensure coverage of all five modes.

Kernel hyperparameters were selected via grid search with 5-fold cross-validation on the training data. The search space consisted of a scaled RBF kernel, $k(\mathrm{x}, \mathrm{x}') = c \cdot \exp\left(-\frac{\|\mathrm{x}-\mathrm{x}'\|^2}{2\ell^2}\right)$, with and without an additive white noise term, evaluated over constant amplitude $c \in \{0.1, 1.0, 10.0\}$, length scale $\ell \in \{0.1, 1.0, 10.0\}$, and noise level $n \in \{0.1, 1.0\}$ for the noise variant. The best-performing kernel configuration was selected based on mean cross-validated accuracy, and its hyperparameters were further refined by the GPC's internal gradient-based marginal likelihood optimizer during training.

The GP classifier was trained on a stratified 80-20 train-test split of the dataset, preserving the class proportions in both subsets. After training, the GP classifier was queried across a dense grid of voltage and pressure combinations spanning the experimental range (2500 V to 6000 V, 0 psi to 5 psi). Per-class precision, recall, and F1-score on the held-out test set are reported in Table 2.

2.3.2 Dynamics prediction with latent transformer

To enable frame-by-frame prediction of electrohydrodynamic printing behaviors, a deep learning architecture was developed that integrates a convolutional autoencoder, transformer-based sequence modeling, and conditional parameter embeddings. The model, illustrated in Fig. 2, operates on sequences of high-speed video frames to predict future nozzle dynamics under specified voltage and pressure conditions. Specifically, the model takes a short window of observed frames as input and predicts the next frame; during inference this prediction is fed back as input to generate the following frame, and this autoregressive rollout is repeated to produce a complete synthetic video sequence. This approach serves in place of computationally intensive physics-based simulations as a data-driven surrogate capable of rapid inference.

The architecture consists of four primary components: (1) a 3D convolutional encoder that extracts spatiotemporal features from input frame sequences, (2) a transformer module that models temporal dependencies in the latent space, (3) a conditional embedding layer that incorporates voltage and pressure parameters, and (4) a 2D deconvolutional decoder that generates pixel-space predictions. A skip connection is used between the encoder and decoder to preserve high-frequency spatial details during frame reconstruction. A latent transformer was chosen due to its ability to model global temporal relationships through self-attention mechanisms, which proved more effective than the sequential processing limitations of Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) networks, which exhibit

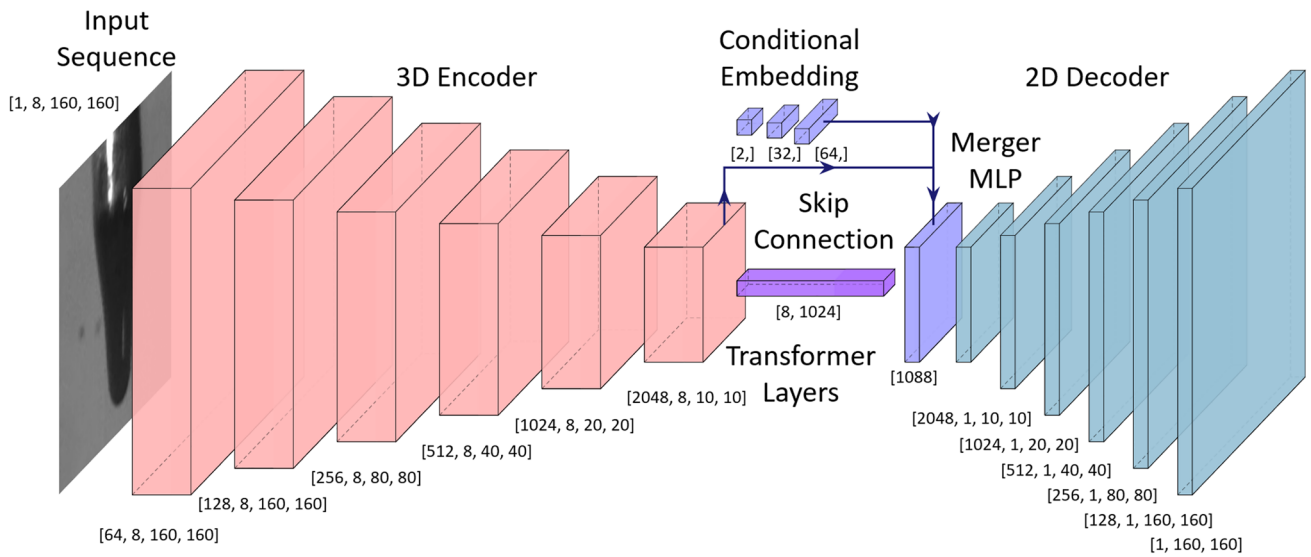


Fig. 2 The deep learning architecture utilized for the autoregressive next-frame prediction class combines a convolutional encoder-decoder to extract structural features and transformer layers to resolve temporal dependencies. A conditional embedding is added to a skip connection, combining the temporal information, structural information, and voltage-pressure context at a merger stage. The final prediction is autoregressively concatenated with the input for the next frame

vanishing gradients, or the local receptive field constraints of purely convolutional approaches [49–53]. Additionally, transformer-based architectures have shown state-of-the-art performance in vision-related tasks [54].

Given an input sequence of T grayscale frames $\{\mathrm{I}_t \in \mathbb{R}^{H\times W}\}_{t=1}^{T}$, the 3D encoder first processes the spatio-temporal features through a series of strided convolutional layers:

$$\mathrm{z}_t = \mathrm{Encoder3D}(\mathrm{I}_t) \in \mathbb{R}^d \tag{4}$$

where d denotes the latent dimension ($d = 1024$). This produces a sequence of latent vectors $\{\mathrm{z}_t\}$ capturing the evolution of nozzle dynamics.

The transformer module then processes these latent vectors using multi-head self-attention to model long-range temporal dependencies. For N transformer layers with h attention heads, the self-attention mechanism at layer l computes:

$$\mathrm{Attention}(\mathrm{Q},\mathrm{K},\mathrm{V}) = \mathrm{softmax}\left(\frac{\mathrm{QK}^T}{\sqrt{d/h}}\right)\mathrm{V} \tag{5}$$

where Q, K, and V are learned linear projections of the input sequence. The implemented architecture uses $N = 5$ transformer layers with $h = 8$ attention heads.

Voltage and pressure conditions are incorporated through a learned embedding network that maps the 2D parameter space (V, P) to 64 dimensions with a multilayer perceptron (MLP):

$$\mathrm{c} = \mathrm{MLP}([V,P]^T) \in \mathbb{R}^{64} \tag{6}$$

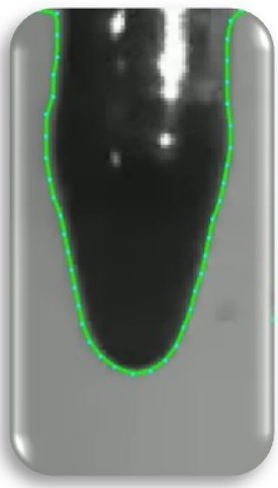


Fig. 3 Representative frame with the outer contour of the fluid meniscus highlighted. Sequential frames of this form constitute the training targets for the deep learning model

This conditional embedding is concatenated with the transformer's output and a skip connection from the 3D encoder before being processed by the merger layer:

$$\hat{\mathrm{z}}_t = \mathrm{MLP}([\mathrm{z}_t^{\mathrm{trans}};\mathrm{c};\mathrm{z}_t^{\mathrm{skip}}]) \tag{7}$$

where $\mathrm{z}_t^{\mathrm{trans}}$ is the transformer output and $\mathrm{z}_t^{\mathrm{skip}}$ is the encoder's latent vector at time t. This combined representation is decoded into predicted frames through a series of transposed convolutions:

$$\hat{\mathrm{I}}_{t+1} = \mathrm{Decoder2D}(\hat{\mathrm{z}}_t) \tag{8}$$

The training dataset consists of sequential grayscale frames drawn from the high-speed video sequences. Each frame captures the outer contour of the fluid meniscus, as illustrated in Fig. 3; this contour defines the time-evolving shape boundary that the model learns to predict. Consecutive frames from the same video form the seed-prediction pairs used during training, so the model learns the temporal progression of meniscus deformation and droplet formation directly from image data.

The model was trained end-to-end on a single NVIDIA A100 GPU using a mean squared error (MSE) objective between predicted and ground truth frames across an 8-frame seed sequence and 1-step prediction horizon. To enhance generalization, the training dataset was augmented with random spatial shifts, horizontal flips, exposure variations, and Gaussian noise applied to input sequences. Overall, 19,000 8-frame seed sequences were generated for training. The architecture implements spectral normalization and layer normalization to stabilize training, along with channel dropout ($p = 0.1$) in both the encoder and decoder modules. Architectural hyperparameters (number of transformer layers, attention heads, latent dimension, and dropout rate) were initially screened via random search with 5-fold cross-validation on a held-out subset of video sequences; due to the computational cost of full model training, many of these parameters were subsequently refined by hand based on validation loss trends. Overfitting was mitigated through data augmentation, dropout, spectral normalization, and early stopping based on held-out validation loss. Dataset sufficiency was confirmed by validation loss convergence on the 19,000-sequence training set.

A controlled comparison of the transformer against LSTM, GRU, and TCN temporal backbone alternatives is provided in the Supplementary Information (Fig. S1), along with a complete summary of the model architecture and training hyperparameters (Table S1).

During inference, the model operates autoregressively: each predicted frame becomes part of the input sequence for subsequent predictions. This enables simulation of extended

Table 1 Summary of experimental, analysis, and machine learning techniques employed in this study, with their respective roles in the overall framework

Technique	Category	Role in this study	Section
EHD jet printing	Experimental	Representative AM process studied. Electric field drives fluid deposition through a Taylor cone across five regimes (cone-jetting, micro-dripping, dripping, unstable, no-print) as a function of voltage and pressure	Sections 1 and 2.1
High-speed photography (15,000 fps, 4.43 μm/px)	Experimental	Captures sub-millisecond printing dynamics invisible at conventional frame rates. Provides the video dataset for all downstream analyses	Sections 2.1 and 3.1
Video sequence tracking (Xcitex ProAnalyst)	Analysis	Tracks droplet frontier frame-by-frame in dripping and micro-dripping sequences; extracts jet breakage velocities as GP regression training data	Sections 2.2 and 3.2
Gaussian Process (GP) classifier	Machine Learning	Classifies voltage–pressure inputs into five printing regimes with uncertainty quantification; constructs the EHD process map (92% accuracy)	Sections 2.3.1 and 3.2
Gaussian Process (GP) regressor	Machine Learning	Predicts jet breakage velocity as a function of voltage and pressure; generates a quantitative velocity process map ($R^2 = 0.955$)	Sections 2.3.1 and 3.2
Deep learning surrogate (3D Conv. Encoder + Transformer + 2D Decoder)	Machine Learning	Autoregressively predicts frame-by-frame droplet dynamics conditioned on voltage and pressure; computationally efficient CFD surrogate ($R^2 \geq 0.89$)	Sections 2.3 and 3.4

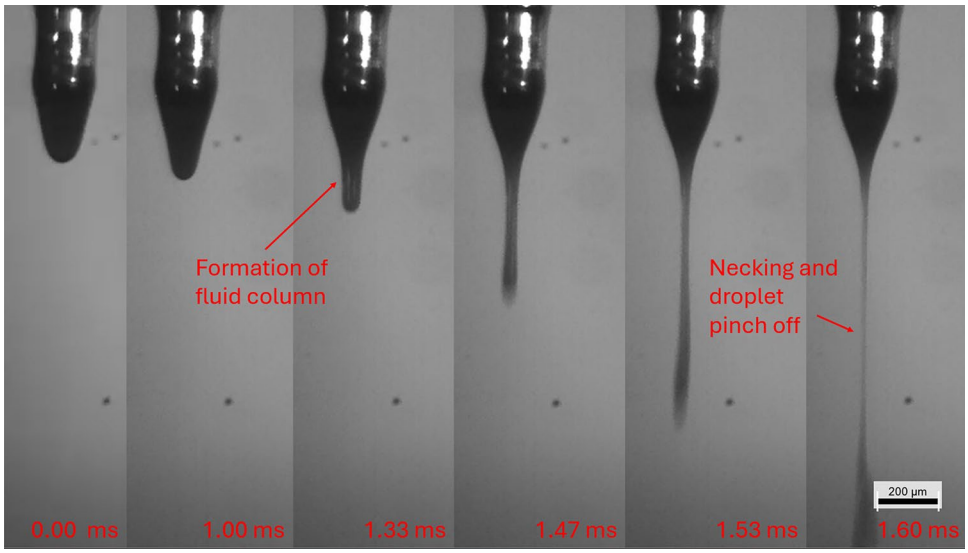


Fig. 4 High-speed photography sequence of micro-dripping EHD printing regime showing periodic formation of small, uniformly-sized droplets through controlled meniscus elongation and pinch-off

temporal dynamics while maintaining dimensional consistency through the architecture. Quantitative validation demonstrated strong agreement with ground truth dynamics predictions (see Sect. 3.4). The integration of conditional

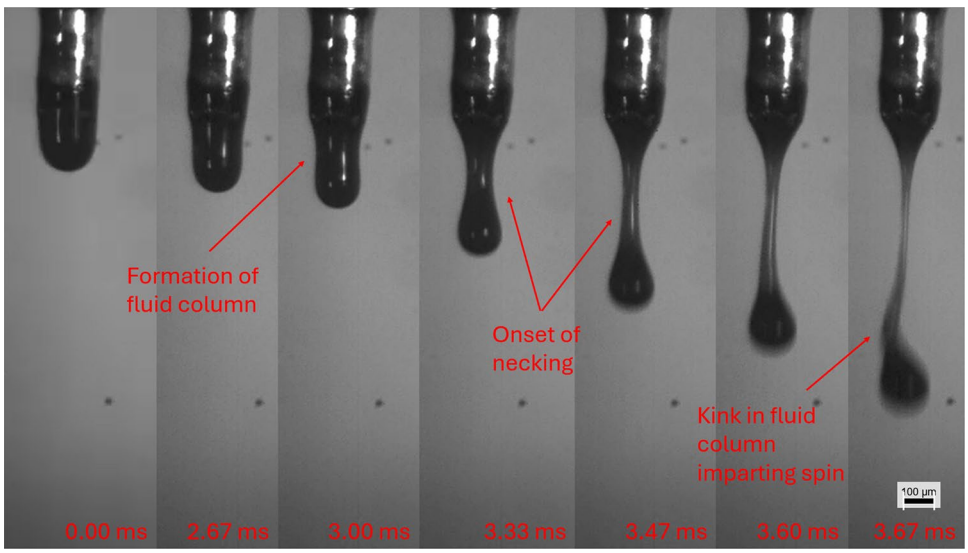


Fig. 5 High-speed photography sequence of dripping EHD printing regime demonstrating larger, more irregular droplet formation with pronounced necking and detachment

embeddings allows the model to generalize across the voltage-pressure parameter space, enabling regime-specific dynamics prediction guided by the process map developed in Sect. 3.2.

Table 1 summarizes the experimental, analysis, and machine learning techniques employed in this study.

3 Results

3.1 Electrohydrodynamic phenomena under high-speed video

3.1.1 Stable micro-dripping, dripping, and cone jet formations

The high-speed imaging system was essential for capturing EHD printing dynamics that occur on sub-millisecond timescales and are invisible to still photography or at typical video frame rates. These rapid phenomena, including droplet formation, jet breakage, and cone dynamics, directly determine print quality but cannot be resolved without sufficient temporal resolution. Video sequences were recorded to capture these transient events across the experimental parameter space. Analysis of the captured sequences revealed distinct behavioral patterns that could be categorized into stable and unstable printing regimes. Figures 4, 5 and 6 show the three major stable printing regimes exhibited by glycerin in the voltage and pressure ranges considered.

The micro-dripping mode, illustrated in Fig. 4, is characterized by periodic ejection of small droplets from the tip of the cone. The regime demonstrates a relatively consistent droplet formation cycle. Droplets are moderately spaced with clear intervals between ejection events. This behavior typically arises under moderate voltage and pressure conditions, particularly in the range of 3000–4600 V with pressures typically below 4 psi, where electrostatic forces are sufficient to overcome surface tension and viscoelastic force but do not yet generate a continuous filament. The frames reveal how the fluid meniscus elongates and begins to pinch off a droplet. The small droplet size and controlled periodicity is desirable in applications requiring high spatial resolution. From a physical standpoint, the stable electrostatic balance enables intermittent droplet detachment without destabilizing the Taylor cone structure.

Figure 5 depicts the dripping mode, in which the nozzle emits relatively large droplets at less consistent frequencies and with more chaotic motion compared to microdripping. This regime occurs at elevated voltages, typically above 4600 V, across a wide pressure range. As seen in the high-speed image sequence, the meniscus at the nozzle tip extends rapidly and bulges into a sizable droplet, causing the cone to rapidly neck, resulting in pinch-off of a large amount of material. The cycle is less regular than in microdripping, indicating the dominance of strong electrostatic forces that destabilize the fluid interface. The substantial momentum imparted to each droplet results in less spatial precision and potential inaccuracies in deposition. The high-speed photography shows that the droplet neck typically distorts and imparts some angular momentum before breaking. This results in reduced control over droplet size, timing, and location.

The cone jetting regime shown in Fig. 6 represents another stable and precise mode for EHD printing. A steady Taylor cone forms at the nozzle tip from which a continuous, thin jet of filament is emitted. Note that this jetting behavior resulted in a very thin jet that was observed on the substrate but not captured by the camera. This mode is typically achieved at lower voltage (below 4000 V) and higher pressure combinations (above 2.5 psi), conditions that

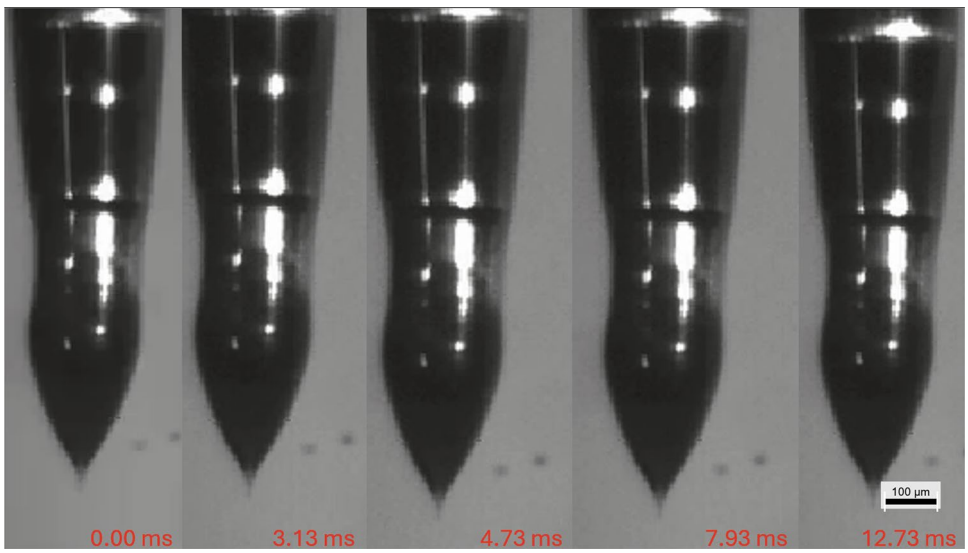


Fig. 6 High-speed photography sequence of cone jetting EHD printing regime exhibiting stable Taylor cone formation with continuous filament ejection, maintaining consistent cone geometry across the 12.73 ms observation period

promote a stable equilibrium between the electrostatic force and the pressure-driven flow rate. In the image sequence, the high-speed video captures the maintenance of a sharply pointed cone with minimal deformation across frames, indicating a highly stable jetting process. This continuous jet allows for fine, uninterrupted deposition, making cone jetting ideal for high-resolution applications such as micropatterning and electronics fabrication. The frames also highlight the uniformity of the jet and absence of large, discrete droplet formations seen in the other modes. The temporal stability captured through high-speed imaging confirms that this regime maintains consistent cone geometry over time, which would be impossible to verify through conventional observation methods that lack sufficient temporal resolution to distinguish true stability from rapid fluctuations.

3.1.2 Jet breakage

Figure 7 presents a sequential high-speed photographic capture of jet breakage in EHD printing, which refers to the process by which the fluid meniscus forms into a Taylor cone and subsequently breaks into discrete droplets.

This sequence highlights the transformation from a stretched fluid column into a separated droplet through necking and thinning. Initially, the jet elongates steadily, but due to a combination of electrostatic forces, viscous drag, and surface tension, the fluid develops a distortion, above which necking occurs. This neck constricts until it pinches off into a clean droplet that separates from the fluid column. The following frames show the recoil of the remaining jet and the free-fall of the newly formed droplet, which for the dripping mode, exhibits some angular momentum with a tail. Understanding jet breakage is essential in EHD printing because it governs the jetting frequency, size, and trajectory of the droplets that are deposited onto the substrate. These rapid dynamics occur over timescales of hundreds of microseconds and would be impossible to resolve using conventional imaging methods or real-time observation. The detailed visualization of necking progression and droplet detachment mechanisms provides critical insights for process optimization that cannot be obtained through post-process part analysis or steady-state modeling approaches.

3.1.3 Unstable formations

High-speed photography was used to capture unstable printing regimes, as well. Understanding unstable jetting regimes is critical for ensuring precision and repeatability in EHD printing processes. These regimes represent transitional or failure states where printing behavior becomes unpredictable, often leading to defects like misaligned or multi-sized droplets, satellite droplets, inconsistent filament widths, or unintended material deposition. By identifying the conditions under which these instabilities arise and characterizing their dynamic behavior, safe operating windows can be accurately defined, process maps can be improved, and real-time control strategies can be developed to avoid or mitigate such behaviors.

Figure 8 illustrates unstable cone jetting, a transitional and temporally fluctuating mode that occurs between cone jetting and either dripping or microdripping behaviors. In this regime, the fluid meniscus initially forms a Taylor cone and begins to emit a jet, but the jetting behavior is inconsistent over time. As seen in the high-speed video frames, the system alternates between periods of relatively steady jetting and moments of collapse or reformation of the cone. This instability typically arises when the applied voltage and pressure are not sufficient to maintain continuous, stable emission but are high enough to intermittently trigger jet initiation if given enough time. The result is a semi-periodic

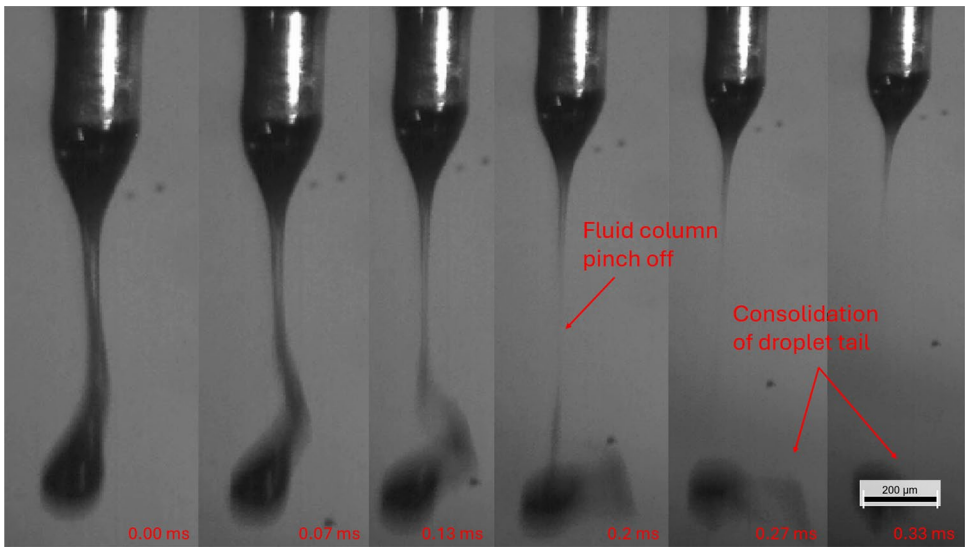


Fig. 7 High-speed photography sequence of jet breakage in EHD dripping mode, showing the complete droplet formation cycle over 0.33 ms. The sequence captures meniscus elongation, neck formation and constriction, droplet pinch-off, and subsequent tail retraction and consolidation with the main droplet body, demonstrating the complex fluid dynamics governing droplet detachment in EHD printing

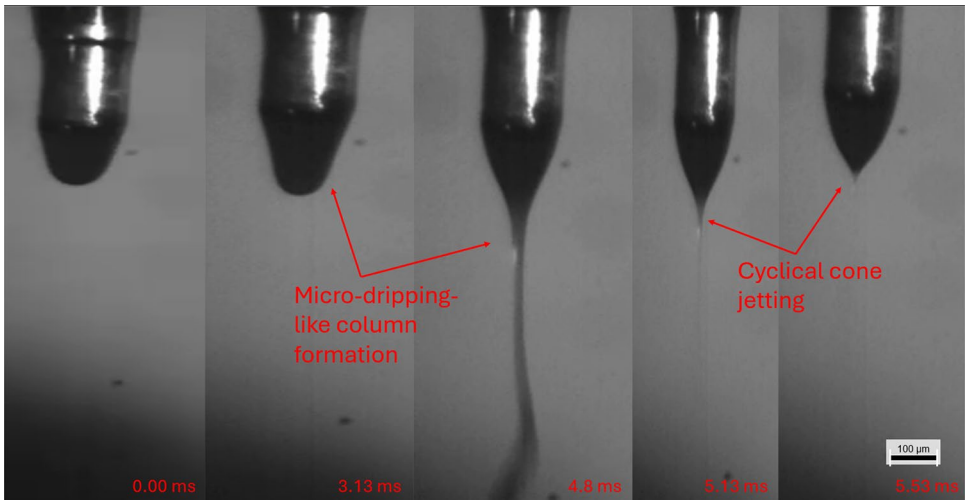


Fig. 8 High-speed photography of unstable cone jetting EHD printing regime showing transitional behavior between stable modes, with intermittent jet formation and collapse over a 5.53 ms sequence, demonstrating the temporal fluctuations that occur in boundary regions of the parameter space

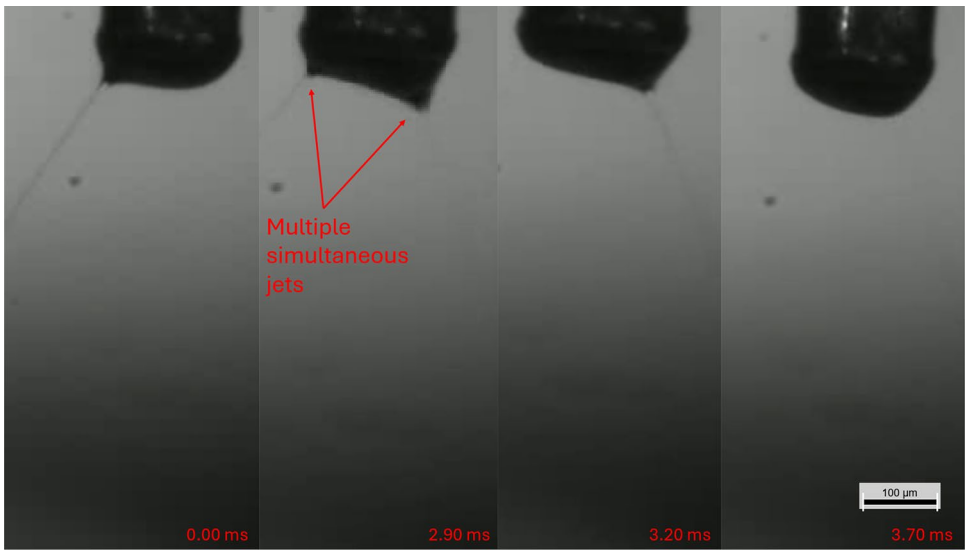


Fig. 9 High-speed photography of multi-jetting EHD printing regime exhibiting simultaneous ejection of multiple fluid streams at divergent angles from the nozzle tip, representing an undesirable high-voltage instability that compromises printing precision and control

oscillation between modes, with transient dripping or partial cone retraction interrupting the jetting process. Such behavior undermines the spatial and temporal precision required for EHD printing, producing inconsistencies in filament diameter, droplet placement, or line continuity.

In contrast, Fig. 9 captures the multi-jetting regime, a distinctly unstable mode that occurs under high-voltage conditions where the electrostatic stress at the nozzle tip exceeds the critical limit for a single jet to form. Instead of a single, centered Taylor cone producing one coherent jet, the fluid interface becomes unstable enough to support the emergence of multiple emission points. These can occur at the periphery of the meniscus or at asymmetric positions along the nozzle edge. The frames show two or more simultaneous jets being ejected at divergent angles, accompanied by irregular cone morphologies. Multi-jetting poses severe challenges for controlled deposition because it produces unpredictable droplet paths and inconsistent material coverage. Understanding this regime is essential for delineating the upper bounds of operational voltages in EHD printing and for preventing structural defects or misalignment in patterned features. High-speed imaging of multi-jetting events provides the resolution needed to capture the rapid onset and evolution of these phenomena, informing both empirical analysis and future feedback control systems aimed at avoiding this mode altogether.

3.2 Printing mode process map

The development of a machine learning-guided process map that captures the relationship between operating conditions and the resultant electrohydrodynamic printing modes will provide a critical framework for process and parameter selections. The process map serves as a data-driven surrogate for traditional experimentation and simulation offering several practical benefits. First, it enables users to identify process parameters that are likely to produce desirable printing modes without requiring extensive trial-and-error

experimentation. Second, it highlights boundaries, mode transitions, and instabilities that should be avoided to ensure consistent print quality. Lastly, this process map forms a foundational layer upon which real-time process control systems or autonomous experimentation platforms can be constructed.

In the process map, the boundaries between the print modes are depicted, and the uncertainty surrounding the boundaries give a measure of the boundary width (i.e., the region of ambiguity between print modes). In this work, the processing map specifically relates voltage and pressure, but the same methodology can be applied to any combination of processing parameters, including printer settings, material properties, and environmental conditions. The print modes observed include cone-jetting, micro-dripping, dripping, unstable behavior, and no-print conditions.

To construct the map (Fig. 10), the Gaussian Process classifier discussed in Sect. 2.3 was trained on the print mode label derived from the 123 high-speed video sequences. The classifier boundaries divide regions where distinct modes dominate, providing a visual guide for selecting operating conditions tailored to specific print outcomes. For each grid point, the model produced a predictive distribution over the five printing modes. The final process map was constructed by assigning each point the mode with the highest predicted probability, while also visualizing the degree of certainty with shading. Regions with high confidence (e.g., well-sampled and consistent behaviors) appear darker, whereas regions of low certainty (e.g., near class boundaries or in undersampled regimes) appear lighter. This allows for clear delineation of printing regions and serves to guide future data acquisition by providing a visualization for targeting regions of high uncertainty. The scattered symbols represent test points, which were held out from the training dataset. An ideal classifier correctly classifies each of the test points. As shown in Fig. 10, the printing modes were divided into distinct regimes with respect to voltage and pressure:

Cone-jetting mode: This mode, which is generally considered to be stable and desirable for high-resolution printing, was found to occur at relatively high pressures (typically above 2.5 psi) and lower voltages (typically below 4000 V). Under these conditions, the fluid meniscus forms a stable Taylor cone and emits a continuous jet. The process map accurately captures this region and suggests that cone-jetting is strongly pressure-driven, with sufficient flow rate required to balance the electric stresses. The voltage requirement to achieve stable cone jetting in glycerin (2.5−4.3 kV) is higher in comparison to previously explored ethanol systems (1.3−3.5 kV), while also requiring some pressure (>2.5 psi). This is expected due to glycerin's high viscosity [55].

Micro-dripping mode: This regime appeared in a distinct mid-range voltage band, approximately between 3600 V and 4600 V. At lower pressures, particularly below 1 psi, micro-dripping was observed for voltages below 3600 V, as well. In micro-dripping, fine droplets are ejected continuously. The model identifies a relatively

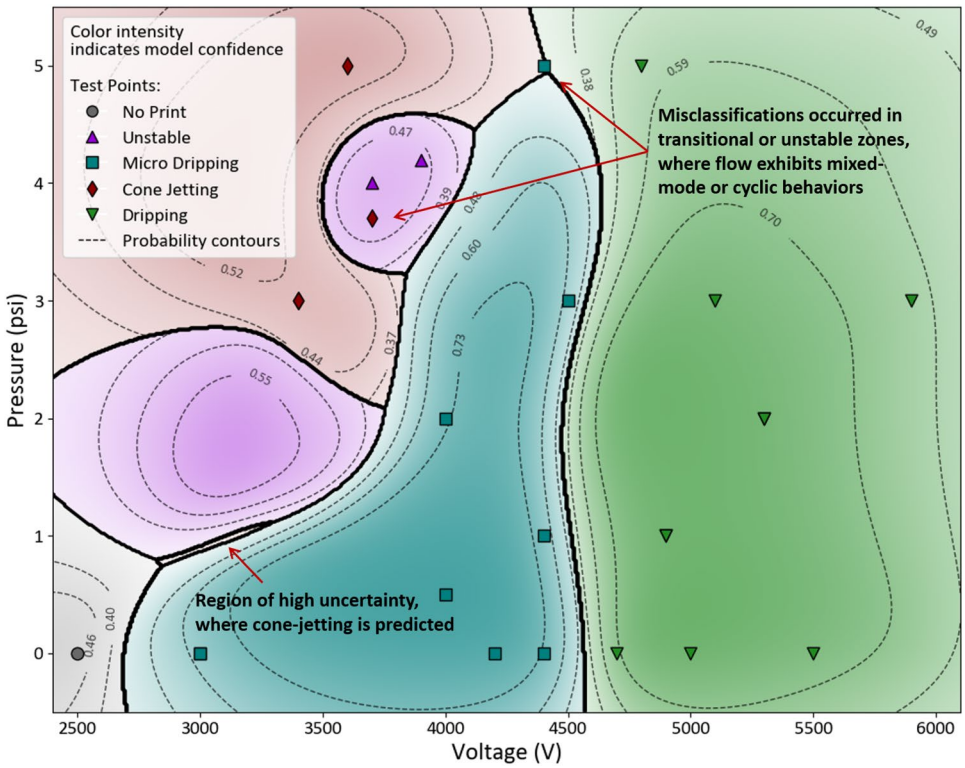


Fig. 10 The relationships between the pressure-voltage control and printing mode are mapped with uncertainty. Red represents the region where cone-jetting is predicted, for combinations of low voltage and high pressure. Teal represents micro-dripping in regions of middle voltage across the pressure range, green represents dripping observed at high voltage, purple represents unstable printing observed between the micro-dripping and cone-jetting regimes, and gray represents no print conditions at low voltage and pressure

wide pressure range for this behavior, suggesting that voltage is the more dominant control parameter in establishing the mode.

Dripping mode: Dripping behavior dominates the higher voltage spectrum (above 4600 V) across the entire pressure range. This regime features large, high-momentum droplets that form and detach rapidly, leading to lower spatial precision. The model reliably detects this region and shows that excessive electric field strength leads to destabilization of the cone and early droplet detachment. The transition to dripping at voltages above 4600 V aligns with established findings that excessive electric field strength destabilizes the Taylor cone geometry and promotes rapid droplet detachment [16].

Unstable regions: Situated between the cone-jetting and micro-dripping modes, this transitional region is marked by unpredictable jetting phenomena, including multi-jetting, side ejections, mode switching, and irregular droplet formation. Videos within this domain often exhibited cyclic or chaotic dynamics, which is expected as the region between micro-dripping and steady cone jetting can yield competing forces resulting in instability [56]. As shown in Fig. 11, the GP classifier showed reduced predictive accuracy within this region, reflecting the intrinsic ambiguity of the underlying physical behavior. Nevertheless, the model provides a useful approximation of the instability envelope by showing high-risk operating points.

No-print region: At the lowest values of both voltage and pressure, the system fails to initiate any electrohydrodynamic phenomena. This corresponds to scenarios where neither sufficient electrostatic force nor material flow exists to form a Taylor cone or eject fluid, as the system fails to meet the threshold voltage [57].

The voltage-dependent mode progression in Fig. 10 reflects the material-specific nature of EHD jetting: EHD behavior is highly sensitive to fluid properties, and glycerin is among the few materials for which both filament and discrete droplet regimes are accessible through pressure-voltage control alone. This progression is consistent with established EHD literature [43].

To evaluate classifier performance, a confusion matrix was generated (Fig. 11). The classifier achieved an overall accuracy of 92%, with misclassifications occurring within the unstable region or along a printing mode boundary shown in Fig. 10. This is expected, as these regions are inherently ambiguous and often exhibit characteristics of multiple modes simultaneously. Despite this, the model shows strong predictive capability across the remainder of the parameter space. Per-class precision, recall, and F1-score are reported in Table 2, providing a more complete picture of classifier performance across all five modes including the minority classes.

The development of this machine learning-guided process map demonstrates how high-speed photography enables mapping of AM parameter spaces that would be

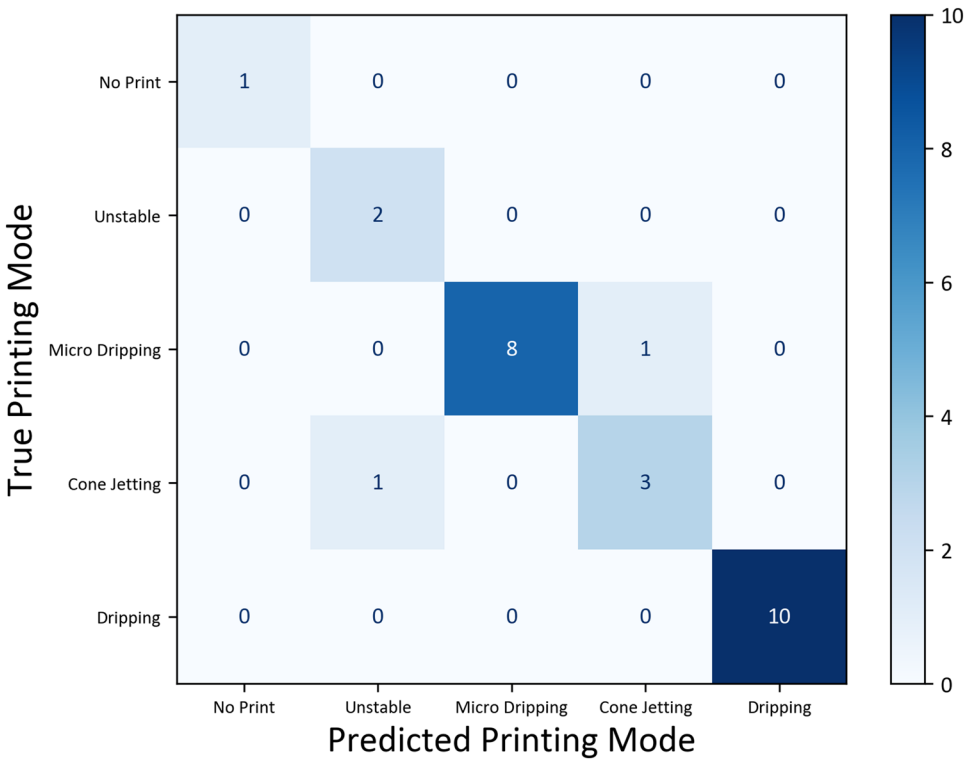


Fig. 11 The confusion matrix shows the performance of the GP classifier in predicting the printing mode (92% accuracy). As shown in the processing map, the two misclassifications were observed in transitional or unstable regimes

Table 2 Per-class precision, recall, and F1-score for the Gaussian Process classifier on the held-out test set (26 samples, stratified 80/20 split)

Printing mode	Precision	Recall	F1-score	Support
No-print	1.000	1.000	1.000	1
Unstable	0.667	1.000	0.800	2
Micro-dripping	1.000	0.889	0.941	9
Cone-jetting	0.750	0.750	0.750	4
Dripping	1.000	1.000	1.000	10
Macro average	0.883	0.928	0.898	26

impossible through conventional approaches. Traditional process map development requires systematic experimentation across parameter combinations, but the rapid timescales of EHD phenomena make real-time classification of printing modes extremely difficult without high-speed imaging. The high capture rate was essential for distinguishing between visually similar but mechanistically different regimes, such as stable cone-jetting versus unstable oscillatory behavior. Manual classification of these modes from conventional imaging would likely be unreliable due to insufficient temporal resolution to capture diagnostic features like droplet formation dynamics and jet stability. The machine learning framework then transformed these high-fidelity observations into a predictive map spanning the entire voltage-pressure parameter space with 92% accuracy. This approach eliminates the need for exhaustive experimental trials at every parameter combination while providing uncertainty quantification that guides future experimentation toward high-uncertainty regions.

3.3 Jet breakage velocities

Understanding and predicting jet breakage velocity represents a critical component in achieving precise control over EHD printing processes. The velocity at which droplets detach from the fluid thread directly influences droplet trajectory, final placement accuracy, and the formation of unwanted satellite droplets that can compromise print quality. In high-resolution additive manufacturing applications, variations in breakage velocity can lead to inconsistent feature dimensions and reduced repeatability, making predictive control of this parameter essential for reliable manufacturing outcomes. Building upon the process mapping framework established in the previous section, this analysis extends the machine learning approach to quantitatively model the relationship between operating parameters and the dynamic behavior of droplet formation.

Jet breakage velocities were measured from the dripping and micro-dripping video sequences using the ProAnalyst tracking procedure described in Sect. 2.2. The velocity of the droplet frontier at the frame immediately preceding pinch-off was recorded for each of the 51 videos spanning the dripping and micro-dripping parameter space, yielding jet breakage velocities used as training data for a Gaussian Process regression model. Gaussian process regression was utilized to create a high-resolution velocity process map, shown in Fig. 12. This map shows the relationship between the EHD printing parameters pressure and voltage and the velocity at which jet breakage occurs. Analysis of Fig. 12 shows a distinct trend, where regions of lower pressure and voltage within the micro-dripping regime exhibit higher velocities. This occurs because micro-droplets tended to elongate further prior to droplet detachment. The fluid column extends significantly before breakage, which causes it to accelerate over a longer travel distance with greater viscous elongation and thinning, allowing time for greater acceleration. The final droplet formations, as shown in Sect. 3.1, are often smaller and more consistently shaped. These characteristics are highly desirable for applications requiring precise material deposition, such as in microelectronics or biofabrication.

Conversely, in the dripping regime, which dominates at higher voltages and wide pressure ranges, jet breakage is far more erratic. The droplets formed are often larger and more irregular, detaching rapidly with shorter elongation periods. The associated velocities exhibit greater variance, likely due to the unstable interplay between high electric field strength and fluid inertia. This mode tended to produce jets that pinch off before achieving significant elongation, resulting in less predictable velocity profiles. The observed inconsistencies in droplet formation are indicative of the more chaotic nature of the dripping mode, where minor fluctuations in operating conditions can lead to significant variations in droplet size and trajectory.

The insights gained through this velocity map are useful as a predictive tool as well as a practical diagnostic framework for interpreting and improving EHD printing behavior and stability. In tandem with the categorical printing mode processing map shown in Fig. 10, the velocity map reveals the underlying gradients and dynamics within each mode, highlighting how changes in pressure and voltage affect the speed and nature of droplet detachment within ostensibly stable regions.

3.4 Machine learning guided prediction of printing dynamics

The ability to simulate frame-by-frame printing dynamics through the combination of high-speed video and machine learning represents the potential for a shift from traditional physics-based modeling to data-driven surrogate approaches. This methodology bypasses the need for explicitly solving complex partial differential equations and learns the underlying multi-physics relationships directly

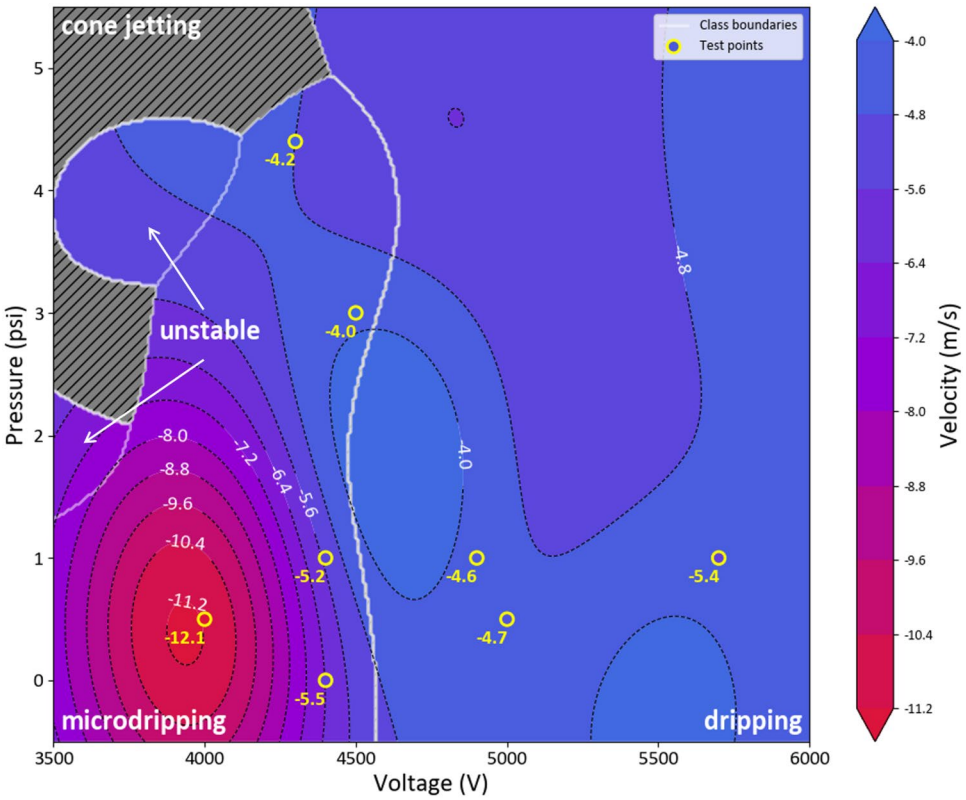
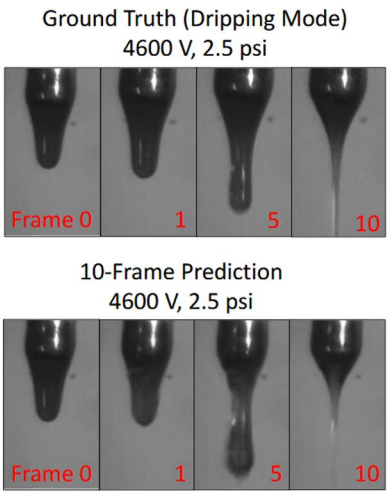
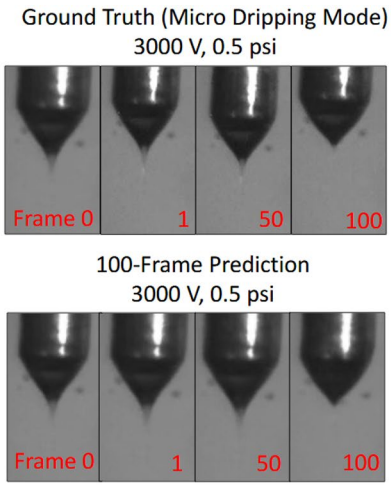


Fig. 12 This process map ($R^2 = 0.955$) shows the relationship between the voltage-pressure control and droplet velocity at the moment the jet breaks. The behavior between the dripping and micro-dripping regimes is distinct: the micro-dripping regime shows significantly high velocities at lower pressures as the meniscus elongated more before breakage than during micro-dripping at high pressures or dripping. Contrastingly, the dripping regime's breakage velocity was approximately constant



(a) 10-frame Dynamics Prediction of Dripping



(b) 100-frame Dynamics Prediction of Micro-dripping

Fig. 13 Comparison of ground truth experimental data (top row) and machine learning model predictions (bottom row) for two EHD printing regimes. **a** Dripping mode at 4600V/2.5psi showing rapid droplet formation and detachment over 10 frames, with the ML model accurately capturing meniscus elongation, necking, and droplet separation dynamics. **b** Micro-dripping mode at 3000V/0.5psi demonstrating slower, more controlled droplet formation over 100 frames, with the ML model successfully predicting the gradual cone evolution and stable droplet release characteristic of this regime

from experimental data. The generated simulations demonstrate how this approach accurately predicts the temporal evolution of droplet formation and detachment dynamics, establishing the viability of using high-speed experimental observations as the foundation for computationally efficient inferential predictive modeling in complex additive manufacturing processes.

Simulations of the printing behavior were generated using the model shown in Sect. 2.3. Figure 13 presents qualitative rollouts of the trained model for both the dripping and micro-dripping modes. The simulations were conducted at voltage and pressure combinations of 4600 V at 2.5 psi for dripping and 3000 V at 0.5 psi for micro-dripping. These values correspond to regions within their respective mode domains as established in the process map (Fig. 10), ensuring the observed dynamics were representative of each printing regime. Dripping and micro-dripping were selected because they are the only regimes that produce discrete,

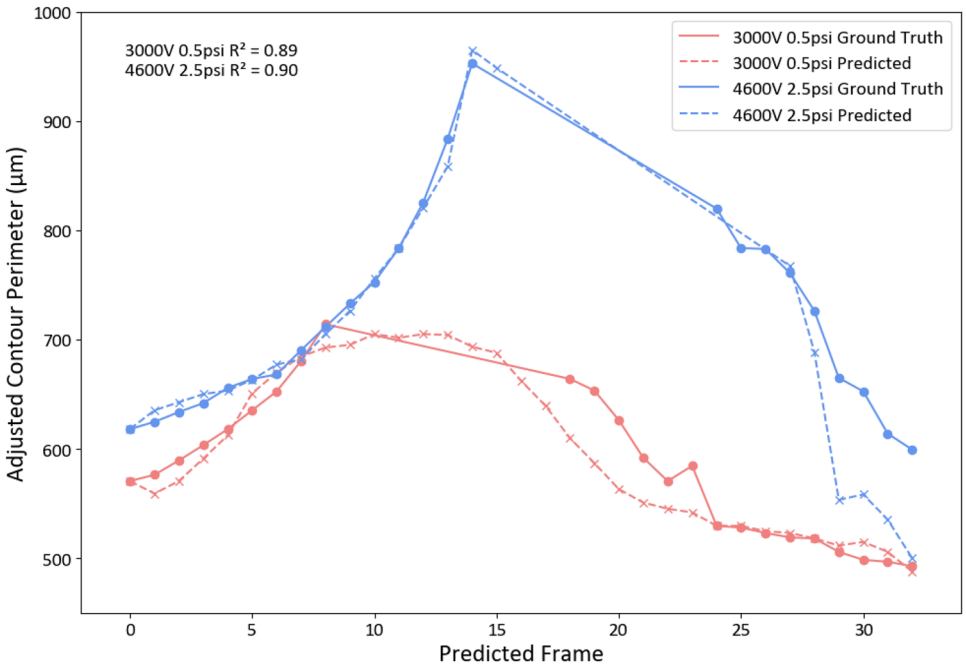


Fig. 14 Quantitative validation of machine learning-predicted electrohydrodynamic printing dynamics through contour perimeter analysis. Contour perimeters (μm) are compared between ground truth experimental data (solid lines) and ML predictions (dashed lines) for micro-dripping at 3000V/0.5psi (red, $R^2 = 0.89$) and dripping at 4600V/2.5psi (blue, $R^2 = 0.90$). The micro-dripping regime shows gradual droplet formation with peak elongation around frame 10, while the dripping regime exhibits rapid dynamics with sharper peak formation around frame 15. Both demonstrate strong correlation between predicted and experimental dynamics, validating the model's ability to capture complex fluid behaviors across different printing modes

repeating droplet events suitable for deterministic frame-by-frame prediction; cone-jetting, unstable, and no-print conditions do not. The two cases were chosen to represent the extremes of this range: dripping at high voltage and pressure, and micro-dripping at low voltage and pressure.

As shown in Fig. 13a, the model successfully predicts the full dripping cycle across a 10-frame rollout, accurately capturing the meniscus elongation, necking, and eventual detachment of the droplet. Although some minor deviations are observed in edge sharpness, the overall morphology and progression of the ejection event align well with the ground truth video frames. The generated sequence reflected the higher kinetic energy and shorter timescales characteristic of the dripping mode.

Figure 13b illustrates the results for the slower micro-dripping regime over a 100-frame sequence. The model is able to resolve the gradual growth and release of smaller droplets, with consistent agreement in terms of meniscus deformation, cone shape, and detachment characteristics. The longer horizon prediction demonstrates the stability and generalization capabilities of the model over extended temporal sequences, which is critical for applications involving continuous process monitoring, such as real-time feedback control.

To quantitatively assess the fidelity of the machine learning predictions against experimental observations, a contour-based comparison approach was adopted for the two regimes shown in Fig. 13. The metric used for comparison was the adjusted contour perimeter, a geometric descriptor representing the total length of the fluid interface shown in each frame. This metric captures both the size and shape of the meniscus and evolving droplet, serving as a robust, physics-relevant indicator of the dynamic behavior at the nozzle tip. The contour perimeter is particularly valuable because it directly reflects the morphological changes occurring during droplet formation and detachment, providing a quantitative measure of the complex fluid deformation processes that characterize different printing regimes.

Figure 14 plots the adjusted contour perimeter (in micrometers) as a function of predicted frame number for both ground truth (solid lines) and ML-predicted sequences (dashed lines). The red curves correspond to the micro-dripping condition, while the blue curves represent the dripping condition. The predicted curves closely track the ground truth profiles across the entire frame range for both printing modes.

In the micro-dripping regime (3000 V, 0.5 psi), the adjusted contour perimeter of the ground truth data exhibits an increase from frame 0 to frame 10, reflecting the extension of the droplet. As the droplet neck thins, the tracking algorithm temporarily loses the contour due to reduced visual resolution and diminished contrast near the pinch-off point. Consequently, the observed "decay" in perimeter after frame 10 does not correspond to the retraction of the meniscus but rather to the loss of tracking fidelity. In the actual video, the meniscus does not begin to return until approximately frame 100. The downward slope observed beyond frame 24, therefore, can be interpreted as a net flow of material out of the nozzle. The ML-predicted perimeter mirrors the ground truth behavior, capturing the overall growth and decay dynamics and final meniscus position with minimal deviations occurring when droplet detachment occurred

sooner than in the ground truth data. The correlation coefficient for this regime is $R^2 = 0.89$, indicating strong agreement between the prediction and experiment despite the longer temporal horizon.

For the dripping regime (4600 V, 2.5 psi), the dynamics are faster and more pronounced. The perimeter rises sharply, reaching a peak near frame 15, before declining rapidly as the droplet detaches and the interface recedes. The ML model accurately predicts both the timing and magnitude of this sharp transition, although some small discrepancies are noted during the decay phase where the predicted final meniscus location is further receded than in the ground truth video. Nevertheless, the prediction remains correlated with the ground truth, with an $R^2 = 0.90$, validating the model's ability to handle rapid dynamics in the dripping mode.

These results demonstrate the ML model's capacity to satisfactorily predict not only visually accurate sequences but also quantitatively faithful geometric features across printing regimes. Some of the error in the predicted contour perimeter results from noise in the perimeter tracking process for both the predicted and ground-truth videos. The prediction accuracy, however, could still be improved. It is expected that additional amounts of training data will improve overall predictive accuracy. Additionally, architectural improvements utilizing the cutting-edge of computer vision research may improve the next-frame prediction performance. However, the correlation is considered to be strong between predicted and true contour perimeter, indicating that the model successfully learned key aspects of the physics governing cone formation, droplet inflation, and detachment. Importantly, the model achieves this without explicit computation of the physical laws during inference, relying solely on learned spatiotemporal features from relatively limited high-speed video data.

4 Discussion and conclusions

This study presents a data-driven framework for understanding the dynamic behaviors of the EHD printing process through the integration of high-speed photography and machine learning. The approach addresses long-standing challenges associated with time-consuming and costly optimization, simulation, and process control by leveraging temporally resolved video data to train models capable of classification and simulation of printing phenomena.

First, a machine learning-based process map was developed to classify operating conditions into printing regimes, including cone-jetting, micro-dripping, dripping, unstable, and no-print modes. Using a Gaussian Process classifier trained on labeled high-speed video data, the process map demonstrated high accuracy (92%) and effectively identified both optimal and unstable regions in the voltage-pressure parameter space. This probabilistic framework provides additive manufacturing operators with a predictive tool for selecting print parameters that optimize stability and desired behavior without the need for exhaustive experimentation.

Second, Gaussian Process regression achieved high predictive accuracy (R^2 = 0.955) in determining the velocity of droplets at jet breakage, revealing distinct regime-dependent trends, where micro-dripping conditions at lower pressures and voltages exhibited significantly higher breakage velocities due to extended meniscus elongation prior to detachment, while the dripping regime demonstrated more consistent velocity profiles across the parameter space. This velocity mapping capability provides manufacturers with quantitative understanding of droplet dynamics that was previously accessible only through expensive computational fluid dynamics simulations requiring extensive computation time.

Third, a deep learning model was trained to simulate the frame-by-frame evolution of droplet formation. This model accurately predicted both rapid and slow printing behaviors, demonstrating its adaptability across different electrohydrodynamic printing modes. Qualitative analysis of rollouts for dripping and micro-dripping closely mirrored ground truth sequences, while quantitative validation via contour perimeter tracking confirmed strong agreement in temporal dynamics. The predictive model successfully captured both the macro-scale features of droplet evolution and the nuanced meniscus dynamics, achieving R^2 values of 0.90 and 0.89 for the dripping and micro-dripping regimes, respectively. The computational efficiency of the proposed framework represents a significant advancement over traditional physics-based simulation approaches, though opportunities for optimization remain. The deep learning model requires approximately 8 h of training time on an A100 GPU to achieve convergence, after which inference becomes remarkably efficient, generating 200-frame dynamic rollouts in approximately 8 s, which is orders of magnitude faster than equivalent computational fluid dynamics simulations.

However, the current architecture is memory-intensive due to the combination of 3D convolutional encoding, transformer-based temporal modeling, and high-resolution image processing, limiting scalability for extended sequences or real-time deployment on resource-constrained hardware. Several immediate computational improvements exist that could improve the feasibility of this methodology on lower-end GPUs. These improvements could include model pruning techniques to reduce the transformer layer complexity, quantization methods to decrease memory footprint, and more efficient 3D convolution alternatives such as separable or depth-wise convolutions [58–60]. Additionally, the inherently parallel nature of frame prediction tasks presents

opportunities for distributed inference across multiple GPUs, potentially enabling real-time processing of multiple nozzle systems simultaneously for large-scale manufacturing applications [61].

A further limitation of the current framework is the scope of the training data, which is restricted to a single fluid, nozzle geometry, and lighting configuration. Expanding the dataset to encompass varied materials, nozzle-based AM systems, and imaging conditions would improve model generalizability and reduce the need for process-specific retraining. Such an effort could form the basis of a foundation model for high-speed video-guided additive manufacturing, enabling rapid adaptation across diverse printing processes with minimal additional data collection after initial pretraining.

Overall, this work demonstrates a pathway toward characterization and predictive control of EHD printing by establishing high-speed imaging-informed machine learning as a viable surrogate for computationally intensive traditional simulation approaches, with broader implications for complex AM processes. The synergy between these technologies was essential. High-speed photography captured the sub-millisecond EHD phenomena that govern print quality, while machine learning frameworks transformed this visual data into predictive tools. High-speed imaging alone would yield only qualitative observations of dynamic behaviors invisible to conventional methods, while machine learning algorithms without rich spatiotemporal input would lack the physical grounding necessary to accurately model the complex multi-physics behavior inherent in additive manufacturing.

This synergistic approach represents a meaningful step toward replacing empirical, trial-and-error parameter selection with predictive, data-driven process guidance, enabling systematic exploration of complex parameter spaces without prohibitive computational overhead. By enabling rapid identification of stable operating regimes and accurate prediction of dynamic behaviors, this framework provides new capabilities for parameter optimization and predictive modeling in EHD printing. Rather than relying on extensive trial-and-error experimentation or computationally expensive simulations, operators can more systematically explore complex parameter spaces through data-driven process maps and virtual experimentation. This capability is particularly valuable for precision manufacturing applications where fine control over complex multi-physics phenomena (e.g., droplet formation or jetting behavior) directly impacts final part quality and dimensional accuracy. The ability to predict frame-by-frame printing dynamics opens possibilities for predictive process monitoring and control strategies, potentially enabling AM systems to automatically detect and correct instabilities, such as keyholing in powder bed fusion, before they compromise part integrity. The demonstrated methodology provides a foundation for similar data-driven approaches in other AM process categories, though generalization of conclusions across fundamentally different process mechanisms warrants further process-specific validation and data collection.

Potential extensions of this work include expanding the parameter space to incorporate additional variables such as nozzle-to-substrate distance, environmental conditions, and material parameters in EHD printing. Including these parameters will improve the generalizability of the models across printer setups and materials, as well as provide crucial insight into the impacts of these various parameters on printing outcomes. In the broader context of additive manufacturing, high-speed imaging-informed machine learning represents a promising avenue for extension to other AM technologies, from laser powder bed fusion spatter dynamics and melt pool behavior to fused deposition modeling extrudate flow and material jet printing processes, though such applications would require process-specific data collection, model adaptation, and validation. Additionally, incorporating physics-informed machine learning frameworks (e.g., Neural ODEs or PINNs) could further enhance prediction accuracy and reduce data requirements by enforcing physics constraints dependent on the specific process within the model architecture [62–64]. Lastly, integration with real-time control systems will enable autonomous process adjustment, transforming predictive models into active feedback mechanisms for stability enforcement and quality assurance in scaled applications [65, 66]. By bridging empirical observations through high-speed photography with data-driven modeling, this research lays the groundwork for scalable, intelligent EHD printing systems that are not only responsive to the complexities of the underlying dynamics, but also capable of adapting in real-time to optimize performance across a broad spectrum of operational conditions. The demonstrated ability to perform simulation and prediction on short inferential timescales enables improved process understanding and virtual experimentation capabilities within EHD printing, contributing toward the longer-term advancement of a vision-based, data-driven predictive paradigm for AM processes where complex multi-physics behaviors can be anticipated and understood without expensive computational overhead.

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Data availability The dataset generated during this project is available at: https://doi.org/10.7910/DVN/WYWLRN

Declarations

Conflict of interest The authors have no financial conflict of interest to disclose.

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